

The Role of Physiological Parameters in the Presence and Severity of Coronary Heart Disease

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1 Introduction

Coronary heart disease (CHD) occurs when the coronary arteries become narrowed, commonly due to plaque buildup, reducing the delivery of oxygen-rich blood to the heart muscle [1]. Clinical measurements may provide information about the presence of CHD, although individual traits may differ in how strongly they are associated with disease.

This project investigated five continuous clinical traits from the Cleveland Heart Disease dataset [2, 3] to answer two questions: which trait best distinguishes patients with and without CHD, and which trait has the strongest relationship with increasing CHD category? The dataset classifies CHD using the variable `num`, ranging from 0 to 4. For this analysis, increasing category was assumed to reflect increasing severity or complexity of CHD, allowing the categories to be treated as ordered. In doing so, it is also a goal for the code to be fairly easiler converted for use with similar disease investagations where data includes broad physiological testing of patients and ordinal disease presence data.

These questions examine different aspects of the data. The first considers whether an individual trait can discriminate between the presence and absence of CHD, while the second considers whether that trait changes systematically as CHD category increases. Logistic regression and receiver operating characteristic analysis were used to investigate CHD presence, while Spearman rank correlation was used to examine relationships with increasing CHD category.

2 Data Sample and Analysis

The Cleveland Heart Disease dataset contains 303 patient observations and 14 attributes [2, 3]. Five continuous traits were selected for analysis: age, resting blood pressure (RBP), serum cholesterol, maximum heart rate achieved, and ST depression induced by exercise. Missing values represented by `?` were converted to `NA`, and observations missing the trait required for an individual analysis were excluded.

The outcome variable `num` ranges from 0 to 4. A value of 0 represents absence of heart disease, while values 1–4 represent CHD. For the disease-presence analysis, `num` was converted into a binary variable in which 0 represented no CHD and 1 represented CHD.

Each trait was analyzed separately using binomial logistic regression in R [4]. Predicted probabilities from each model were evaluated using receiver operating characteristic (ROC) analysis with the `pROC` package [5]. Area under the ROC curve (AUC) measured how well each trait distinguished patients with and without CHD. An AUC of 0.5 represents no discrimination, while values increasingly above 0.5 indicate greater discriminatory ability. A 95% confidence interval was calculated for each AUC to represent uncertainty around the estimate.

For the second analysis, the original CHD categories from 0 to 4 were retained. Spearman correlation was used to measure the strength and direction of the relationship between each trait and increasing CHD category. Positive coefficients indicate that a trait tends to increase with CHD category, while negative coefficients indicate that it tends to decrease. Absolute coefficient values were used when comparing relationship strength.

Uncertainty in the Spearman estimates was assessed using 1,000 bootstrap samples. Patient observations were repeatedly sampled and the Spearman correlation recalculated for each sample. The 2.5th and 97.5th percentiles of the resulting distribution were used as the 95% confidence interval. A fixed random seed (123) was used to ensure reproducibility.

Statistical analyses were performed in R, and figures were produced in Python. The AUC and Spearman plots display 95% confidence intervals as error bars, with AUC = 0.5 and Spearman $\rho = 0$ included as reference values. Maximum heart rate and ST depression were also plotted at the patient level according to CHD category.

3 Results and Discussion

The results of both analyses are summarized in Table 1. Maximum heart rate and ST depression produced the strongest results, although their direction and relative strength differed depending on whether CHD presence or increasing severity category was examined.

Table 1: Predictive ability and association with increasing CHD category for selected clinical traits. Values in brackets represent 95% confidence intervals.

Clinical trait	**AUC (95% CI)**	**Spearman ρ (95% CI)**
Age	0.637 (0.574–0.700)	0.242 (0.140–0.358)
Resting blood pressure	0.574 (0.510–0.639)	0.137 (0.032–0.254)
Serum cholesterol	0.570 (0.505–0.635)	0.109 (−0.013–0.227)
Maximum heart rate	0.745 (0.690–0.801)	−0.444 (−0.530–−0.351)
ST depression	0.735 (0.679–0.792)	0.463 (0.359–0.549)

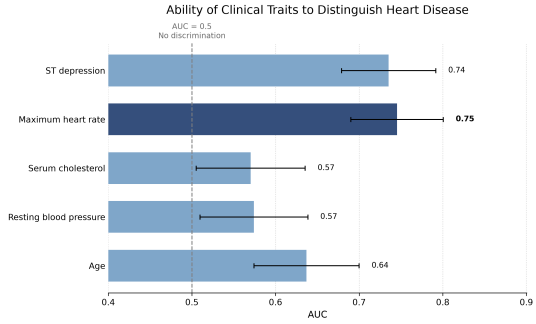
Maximum heart rate had the highest observed AUC, witnessed in Figure 1a below at 0.745 (95% CI 0.690–0.801), followed closely by ST depression at 0.735 (95% CI 0.679–0.792). Both confidence intervals remained above the AUC = 0.5 no-discrimination reference, indicating moderate discriminatory ability above chance within this sample. However, the observed AUC values differed by only 0.010 and their confidence intervals overlapped considerably. This does not provide strong evidence of a meaningful difference in discriminatory ability between the two variables. Age showed weaker discrimination with an AUC of 0.637, while resting blood pressure and serum cholesterol were closer to 0.5, with AUCs of 0.574 and 0.570 respectively.

For increasing CHD category, seen in Figure 1b below, ST depression showed the strongest observed Spearman correlation ($\rho = 0.463$, 95% CI 0.359–0.549), followed closely by maximum heart rate ($\rho = -0.444$, 95% CI −0.530–−0.351). Their similar absolute magnitudes indicate relationships of similar strength, while their signs indicate opposite directions. Greater ST depression tended to occur in higher CHD categories, whereas maximum heart rate tended to decrease. Exercise-induced ST-segment depression is a recognized marker used in the assessment of coronary artery disease (broader term including CHD), although its diagnostic accuracy varies between populations and testing methods [6]. Reduced chronotropic response (appropriate response of heart rate to exercise) during exercise has also been associated with coronary disease and adverse cardiovascular outcomes [7]. Therefore, the negative coefficient for maximum heart rate represents an inverse relationship rather than a weaker one.

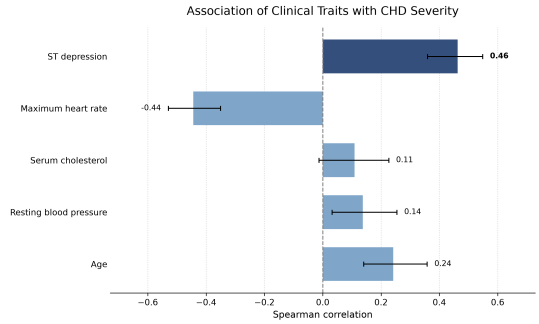
Age and resting blood pressure showed weaker positive associations, while the confidence interval for serum cholesterol (−0.013–0.227) included zero, meaning little or no monotonic relationship is also consistent with the data. This should not be interpreted as evidence that blood pressure or cholesterol are unimportant in coronary disease. Elevated blood pressure and atherogenic cholesterol are established risk factors for atherosclerotic cardiovascular disease [9]. Instead, the present results suggest that single measurements of these variables were relatively weak at distinguishing existing CHD within this particular sample.

The patient-level plot in Figure 1c provides additional context for the two strongest traits. Higher CHD categories generally occur with greater ST depression and lower maximum heart rate, consistent with the positive and negative Spearman coefficients. However, considerable overlap remains between categories, showing that neither variable alone can completely separate patients according to disease category.

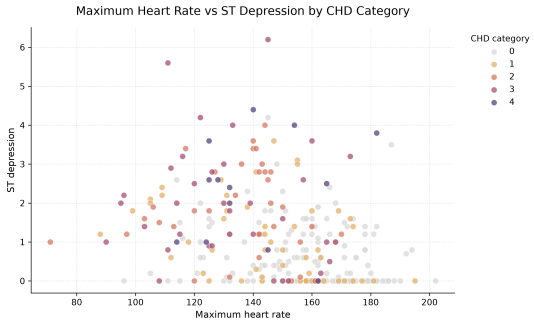
Figure 1: Comparison of physiological traits associated with CHD presence and increasing CHD category.



(a) AUC for the five clinical traits. The dashed line represents no discrimination at AUC = 0.5. Error bars show 95% confidence intervals.



(b) Spearman correlation between each trait and CHD category. Error bars show 95% confidence intervals.



(c) Maximum heart rate and ST depression according to CHD category.

Overall, maximum heart rate and ST depression were the two most informative traits in this analysis. Their performance was similar, but each was strongest for a different outcome: maximum heart rate for distinguishing CHD presence and ST depression for association with increasing CHD category.

Several limitations should be considered when interpreting these findings. Each logistic regression model included only one clinical trait, so possible combined effects or interactions between variables were not examined. In addition, the same dataset was used to fit and evaluate the models, meaning the AUC values describe discrimination within this sample rather than performance in an independent population. The analysis also assumes that categories 0 to 4 form an ordered progression reflecting increasing severity or complexity of CHD. The Spearman analysis therefore describes trends under this assumption rather than establishing that the categories represent equal or clinically validated increases in severity. Finally, correlation measures association rather than causation, so the results do not demonstrate that changes in these traits directly cause more severe or complex CHD.

A broader aim of the project was to develop a reusable analysis framework rather than code specific to CHD. The variables to be tested are defined in the `traits` vector, so different continuous factors can be analyzed by changing the trait names while keeping the same analysis structure. With an appropriately coded disease outcome, the same workflow can therefore be adapted to compare candidate predictors for other diseases, using AUC to assess discrimination of disease presence and Spearman correlation when an ordered disease category is available. The code is therefore most useful as a screening framework for comparing individual candidate predictors before developing more complex multivariable models.

References

[1] National Heart, Lung, and Blood Institute. *What Is Coronary Heart Disease?* Available at: https://www.nhlbi.nih.gov/health/coronary-heart-disease.

[2] Janosi, A., Steinbrunn, W., Pfisterer, M., and Detrano, R. *Heart Disease Dataset*. UCI Machine Learning Repository, 1989. Available at: https://archive.ics.uci.edu/dataset/45/heart+disease. DOI: 10.24432/C52P4X.

[3] Detrano, R., Janosi, A., Steinbrunn, W., Pfisterer, M., Schmid, J.-J., Sandhu, S., Guppy, K. H., Lee, S., and Froelicher, V. *International Application of a New Probability Algorithm for the Diagnosis of Coronary Artery Disease*. The American Journal of Cardiology, 64(5), 304–310, 1989. DOI: 10.1016/0002-9149(89)90524-9.

[4] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria. Available at: https://www.r-project.org/.

[5] Robin, X., Turck, N., Hainard, A., Tiberti, N., Lisacek, F., Sanchez, J.-C., and Müller, M. *pROC: An Open-Source Package for R and S+ to Analyze and Compare ROC Curves*. BMC Bioinformatics, 12, 77, 2011. Available at: https://doi.org/10.1186/1471-2105-12-77.

[6] Chaudhry, S., Kumar, N., Arena, R., and Verma, S. The evolving role of cardiopulmonary exercise testing in ischemic heart disease: state of the art review. *Current Opinion in Cardiology*, 38(6), 552–572, 2023. doi:10.1097/HCO.0000000000001086.

[7] Carvalho, T., Freitas, O. G. A., Chalela, W. A., Hossri, C. A. C., Milani, M., Buglia, S., Precoma, D. B., et al. Brazilian Guideline for Exercise Test in the Adult Population – 2024. *Arquivos Brasileiros de Cardiologia*, 121(3), e20240110, 2024. doi:10.36660/abc.20240110.

[8] Lauer, M. S., Okin, P. M., Larson, M. G., Evans, J. C., and Levy, D. Impaired heart rate response to graded exercise: prognostic implications of chronotropic incompetence in the Framingham Heart Study. *Circulation*, 93(8), 1520–1526, 1996.

[9] Visseren, F. L. J., Mach, F., Smulders, Y. M., et al. 2021 ESC Guidelines on cardiovascular disease prevention in clinical practice. *European Heart Journal*, 42(34), 3227–3337, 2021. doi:10.1093/eurheartj/ehab484.